HOMEWORK E.4 · EOG REVIEW · NC.8.F.1 · NC.8.F.3 · NC.8.EE.7 · NC.8.F.4 · NC.8.G.5 · NC.8.F.2 · NC.8.G.9

EOG Review: Functions & slope

Targets: F.1-2, F.3-4, F.3-2, EE.7-3, F.4-4, F.1-6, F.4-6, G.5-6, F.2-7, G.9-9

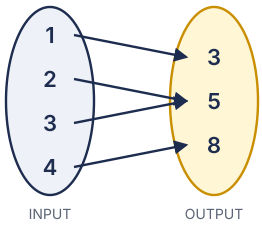
Name _____ Date _____ Block _____

Remember: Function: each input has exactly one output (vertical line test). **Slope** $m = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$. In $y = mx + b$, m = slope (rate of change), b = y-intercept (start value). Example: (1, 2) and (3, 8): $m = \frac{6}{2} = 3$.

SCORE 3.0 Approaching · the basics

Targets: F.1-2, F.3-4, F.3-2, EE.7-3, F.4-4

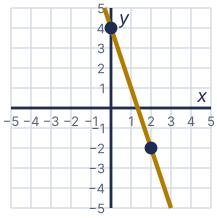
1 Is this mapping a function? F.1-2



Yes No

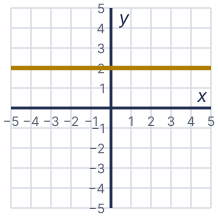
Why? _____

2 Find the slope and the y-intercept. F.3-4



_____

3 What kind of slope does this line have? F.3-2



Positive Negative Zero Undefined

4 Solve: $6x - 11 = 2x + 9$ REVIEW 3.3 EE.7-3

_____

_____

5 A taxi charges \$3 to start plus \$2 per mile. Which equation shows the cost y for x miles? F.4-4

(A) $y = 2x + 3$ (B) $y = 3x + 2$ (C) $y = 5x$ (D) $y = 2x - 3$

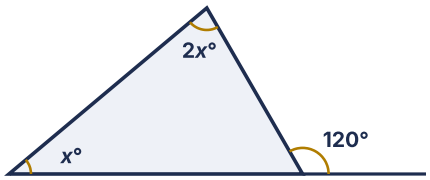
SCORE 4.0 Proficient · on target

Targets: F.1-6, F.4-6, G.5-6

6 If $f(x) = -3x + 4$, find $f(-2)$. F.1-6

7 Write the equation of the line through $(-1, 6)$ and $(3, -2)$. F.4-6

8 Find x. REVIEW 4.5 G.5-6



SCORE 5.0 Mastery · go deeper

Targets: F.2-7, G.9-9

9 Plan A is shown in the table (x = months, y = total cost in dollars). Plan B costs $y = 25x + 20$. Which plan costs less for 10 months? Use numbers. F.2-7

x	1	2	3
y	70	90	110

Plan A's equation is _____ . For 10 months, A costs _____ and B costs _____ . So _____ .

10 An ice cream cone has radius 3 cm and height 8 cm. A half-sphere of ice cream with radius 3 cm sits on top. Find the total volume in terms of π. Explain each step. REVIEW 9.3 G.9-9

The cone is _____ . The half-sphere is _____ . The total is _____ .